

$$\binom{n}{r}$$



$$n \times (n-1) \times \dots$$

Counting Principle Permutations & Combinations

Practice Question Set

**Multiplication Principle, Arrangements with Restrictions, and
Combinations**

28 Questions • Fully Worked Solutions

Instructions: A key formulas and method-notes page follows this cover page. Questions are ordered from Easy to Medium to Hard.

Attempt each question first; the full worked solution is on the page immediately following, boxed in light orange. Where the method of gaps also applies, a second alternative solution is given in a blue box.

Key Formulas & Method Notes

Formulas

- **Arrangements of n distinct objects (all of them, in a row):** $n!$
- **Permutations** (order matters — arranging r objects chosen from n):

$${}^nP_r = \frac{n!}{(n-r)!}$$

- **Combinations** (order does not matter — choosing r objects from n):

$$\binom{n}{r} = {}^nC_r = \frac{n!}{r!(n-r)!}$$

“Together” vs. “Apart”: which method to use?

Most restriction questions fall into one of two opposite requirements, and each has its own natural technique.

1. Items must stand together (the “block” method).

Glue the items that must be together into a single block. Arrange this block along with everything else, then multiply by the number of ways to order the items *inside* the block. *Always* use this method when the requirement is togetherness — never the gap method, which only keeps things apart.

2. Items must *not* be together (kept apart) — two competing methods.

(a) **Complement method (“Total – Together”):** count all arrangements, then subtract the arrangements where the restricted items *are* together (using the block method above for that subtracted case). This is quick and safe when there is only *one* pair (or one specific set treated as a single unit) to keep apart.

(b) **Method of gaps:** arrange all the *other*, unrestricted items first — this creates gaps before, between and after them. Then place each of the restricted items into its *own separate* gap, one at a time, so no two of them can ever end up next to each other.

How to tell them apart: once **three or more** specific items must be *mutually* kept apart (no two of them adjacent to one another), the complement method needs full inclusion–exclusion over every pair (and checking every triple, quadruple, ...) that could be together — and it is easy to make an error here (for instance, three specific people being *consecutive as a block* is not the same event as all three being *pairwise adjacent*, since in a block of three, the two end people are never adjacent to each other!). The method of gaps sidesteps this completely: since every restricted item sits alone in its own gap, no two of them can ever be neighbours, however many there are. As a rule of thumb:

- Keeping **one pair** apart → complement method is usually quickest.
- Keeping **three or more** items mutually apart → method of gaps is safer and shorter.

Several questions in this set (marked with a blue **Alternative Solution** box) show both methods side by side on the same question, so you can compare them directly.

Question 1

[Maximum mark: 4]

Calculator

Difficulty: Easy

Find the number of ways in which twelve different baseball cards can be given to Emily, Harry, John and Olivia, if Emily is to receive 5 cards, Harry is to receive 3 cards, John is to receive 3 cards and Olivia is to receive 1 card.

Question 1

Solution

Choose 5 of the 12 cards for Emily, then 3 of the remaining 7 for Harry, then 3 of the remaining 4 for John, and the last card goes to Olivia. By the multiplication principle, the number of ways is

$$\binom{12}{5} \binom{7}{3} \binom{4}{3} \binom{1}{1} = \frac{12!}{5!7!} \times \frac{7!}{3!4!} \times \frac{4!}{3!1!} \times 1 = \frac{12!}{5!3!3!1!}.$$

Evaluating,

$$792 \times 35 \times 4 \times 1 = \boxed{110\,880}.$$

Question 2

[Maximum mark: 4]

Calculator

Difficulty: Easy

Mark has created an online test for his Physics students to practice in class. There are 12 questions to be attempted in random order. Two of the questions require the use of a calculator. Find the number of different ways in which a student can attempt the test and encounter the two calculator-required questions consecutively.

Question 2

Solution

Treat the two calculator-required questions as one block. Together with the remaining 10 questions, this gives 11 items to arrange, and the block's two questions can be ordered internally in $2!$ ways. The number of ways is

$$\begin{aligned} 11! \times 2! &= 39\,916\,800 \times 2. \\ &= \boxed{79\,833\,600 \approx 7.98 \times 10^7}. \end{aligned}$$

Question 3

[Maximum mark: 5]

Calculator

Difficulty: Easy

A team of four is to be chosen from a group of four boys and four girls.

- (a) Find the number of different possible teams that could be chosen. [3]
- (b) Find the number of different possible teams that could be chosen, given that the team must include at least one girl and at least one boy. [2]

Question 3

Solution

(a) We choose 4 players from the $4 + 4 = 8$ available. The number of possible teams is

$$\binom{8}{4} = \boxed{70}.$$

(b) The complement of “at least one girl and at least one boy” is “all boys” or “all girls”. There is exactly $\binom{4}{4} = 1$ all-boy team and $\binom{4}{4} = 1$ all-girl team, so

$$70 - 1 - 1 = \boxed{68}.$$

Question 4

[Maximum mark: 2]

Calculator

Difficulty: Easy

Consider a three-digit code abc , where each of a , b and c is assigned one of the values 1, 2, 3, 4 or 5. Find the total number of possible codes

- (i) assuming that each value can be repeated (for example, 121 or 444); [1]
- (ii) assuming that no value is repeated. [1]

Question 4

Solution

(i) Each of a, b, c can independently be any of the 5 values, so the number of codes is

$$5 \times 5 \times 5 = \boxed{125}.$$

(ii) Now a, b, c must be three *different* values chosen from the 5 available, in order:

$$5 \times 4 \times 3 = \boxed{60}.$$

Question 5

[Maximum mark: 4]

Calculator

Difficulty: Easy

A particular service group at a school has ten members. The group intends to select a committee of seven by first choosing a chairperson, a vice-chairperson and a treasurer, then selecting four additional members to complete the committee. Determine the number of ways in which this committee can be chosen.

[4]

Question 5

Solution

The three named roles (chairperson, vice-chairperson, treasurer) are filled by three *different* people from the 10 members, and since the roles are distinguishable, order matters:

$${}^{10}P_3 = 10 \times 9 \times 8 = 720 \text{ ways.}$$

The remaining 4 (unnamed) committee members are then chosen from the 7 people left over:

$$\binom{7}{4} = 35 \text{ ways.}$$

By the multiplication principle, the total number of ways to choose the committee is

$$720 \times 35 = \boxed{25\,200}.$$

Question 6

[Maximum mark: 4]

Calculator

Difficulty: Medium

Tyler needs to decide the order in which to schedule 11 exams for his school. Two of these exams are Chemistry (1 SL and 1 HL). Find the number of different ways Tyler can schedule the 11 exams given that the two Chemistry subjects must not be consecutive.

Question 6

Solution

Let n be the number of ways to schedule the 11 exams so that the two Chemistry exams are *not* consecutive. Subtract from $11!$ (all schedules) the number of schedules where the two Chemistry exams *are* consecutive: treat the two Chemistry exams as one block, giving 10 items to arrange in $10!$ ways, with the block's two exams orderable in $2!$ ways internally.

$$n = 11! - 10! \times 2! = 39\,916\,800 - 3\,628\,800 \times 2 = 39\,916\,800 - 7\,257\,600.$$

$$n = \boxed{32\,659\,200 \approx 3.27 \times 10^7}.$$

Question 6

Alternative Solution (Method of Gaps)

Arrange the other 9 exams first: $9!$ ways. This creates 10 gaps (before, between and after them). Place the 2 distinct Chemistry exams into 2 *different* gaps, in order:

$$n = 9! \times (10 \times 9) = 362\,880 \times 90.$$

$$n = \boxed{32\,659\,200 \approx 3.27 \times 10^7}.$$

Question 7

[Maximum mark: 6]

Calculator

Difficulty: Medium

A police department has 4 male and 7 female officers. A special group of 5 officers is to be assembled for an undercover operation.

- (a) Determine how many possible groups can be chosen. [2]
- (b) Determine how many groups can be formed consisting of 2 males and 3 females. [2]
- (c) Determine how many groups can be formed consisting of at least one male. [2]

Question 7

Solution

(a) We choose 5 officers from the $4 + 7 = 11$ available. The number of possible groups is

$$\binom{11}{5} = \boxed{462}.$$

(b) We choose 2 officers from the 4 males and 3 officers from the 7 females. The number of such groups is

$$\binom{4}{2} \binom{7}{3} = 6 \times 35 = \boxed{210}.$$

(c) The complement of “at least one male” is “no males” (all 5 officers female):

$$\binom{4}{0} \binom{7}{5} = 1 \times 21 = 21.$$

Hence the number of groups with at least one male is

$$462 - 21 = \boxed{441}.$$

Question 8

[Maximum mark: 6]

Calculator

Difficulty: Medium

A school basketball team of 5 students is selected from 8 boys and 4 girls.

- (a) Determine how many possible teams can be chosen. [2]
- (b) Determine how many teams can be formed consisting of 3 boys and 2 girls. [2]
- (c) Determine how many teams can be formed consisting of at most 3 girls. [2]

Question 8

Solution

(a) We choose 5 students from the $8 + 4 = 12$ available. The number of possible teams is

$$\binom{12}{5} = \boxed{792}.$$

(b) We choose 3 boys from 8 and 2 girls from 4:

$$\binom{8}{3} \binom{4}{2} = 56 \times 6 = \boxed{336}.$$

(c) The complement of “at most 3 girls” is “all 4 girls” (and hence 1 boy):

$$\binom{8}{1} \binom{4}{4} = 8 \times 1 = 8.$$

Hence the number of teams with at most 3 girls is

$$792 - 8 = \boxed{784}.$$

Question 9

[Maximum mark: 6]

Calculator

Difficulty: Medium

Julie works at a book store and has nine books to display on the main shelf of the store. Four of the books are non-fiction and five are fiction. Each book is different. Determine the number of possible ways Julie can line up the nine books on the main shelf, given

- (a) the non-fiction books should stand together; [2]
- (b) the non-fiction books should stand together on either end; [2]
- (c) the non-fiction books should stand together and do not stand on either end. [2]

Question 9

Solution

(a) Let k be the number of arrangements with the 4 non-fiction books together. Treat them as a single block: this block plus the 5 fiction books gives 6 items, arranged in $6!$ ways, with the block itself orderable internally in $4!$ ways.

$$k = 6! \times 4! = 720 \times 24 = \boxed{17\,280}.$$

(b) Let m be the number of arrangements with the non-fiction books together on either end. There are 2 choices of which end the block occupies, and the 5 fiction books fill the rest in $5!$ ways, with the block internally ordered in $4!$ ways.

$$m = 2 \times 5! \times 4! = 2 \times 120 \times 24 = \boxed{5\,760}.$$

(c) Let n be the number of arrangements with the non-fiction books together and **not** on either end. Using parts (a)–(b),

$$n = k - m = 17\,280 - 5\,760 = \boxed{11\,520}.$$

Question 10

[Maximum mark: 6]

Calculator

Difficulty: Medium

Six women and two men wait in a grocery store checkout line. Determine the number of possible ways these customers can stand in line, given that

- (a) the men do not stand together; [2]
- (b) the men do not stand on either end; [2]
- (c) the men do not stand together and do not stand on either end. [2]

Question 10

Solution

(a) Let k be the number of arrangements where the two men do **not** stand together. Subtract, from the total $8!$ arrangements, those where the men *do* stand together (treat them as one block: $7!$ arrangements of 7 items, times $2!$ internal orderings):

$$k = 8! - 2 \times 7! = 40\,320 - 2 \times 5\,040 = \boxed{30\,240}.$$

(b) Let m be the number of arrangements where the men do not stand on either end. Choose which of the 6 women stands on each end (6 then 5 choices), then arrange the remaining 6 customers (2 men + 4 women) in the middle 6 places in $6!$ ways:

$$m = 6 \times 5 \times 6! = 30 \times 720 = \boxed{21\,600}.$$

(c) Let n be the number where the men neither stand together nor on either end. Using the same end-selection (6×5 ways) but now subtracting, from the $6!$ middle arrangements, the $2 \times 5!$ arrangements where the two men *are* adjacent within the middle six places:

$$n = 6 \times 5 \times (6! - 2 \times 5!) = 30 \times (720 - 240) = 30 \times 480 = \boxed{14\,400}.$$

Question 10

Alternative Solution (Method of Gaps)

Parts (a) and (c) can also be solved directly with the method of gaps (part (b) is not a mutual-separation problem, so it keeps its method above).

(a) Arrange the 6 women first: $6!$ ways, creating 7 gaps. Place the 2 distinct men into 2 *different* gaps, in order:

$$k = 6! \times (7 \times 6) = 720 \times 42 = \boxed{30\,240}.$$

(c) Arrange the 6 women first: $6!$ ways, creating 7 gaps, but the 2 *end* gaps would put a man at either end of the line, so only the 5 *interior* gaps are allowed. Place the 2 distinct men into 2 different interior gaps, in order:

$$n = 6! \times (5 \times 4) = 720 \times 20 = \boxed{14\,400}.$$

Question 11

[Maximum mark: 5]

Calculator

Difficulty: Medium

Sophia and Zoe compete in a freestyle swimming race where there are no tied finishes and there is a total of 10 competitors. Find the total number of possible ways in which the ten swimmers can finish if Zoe finishes

- (a) in the position immediately after Sophia; [2]
- (b) in any position after Sophia. [3]

Question 11

Solution

(a) Treat “Sophia immediately followed by Zoe” as one block. This block plus the remaining 8 swimmers gives 9 items to arrange:

$$n = 9! = \boxed{362\,880}.$$

(b) Suppose Sophia finishes in position p (from 1st to 9th, since Zoe must finish after her). If Sophia is in position p , there are $10 - p$ positions after her where Zoe can finish, and the remaining 8 swimmers fill the other 8 places in $8!$ ways. Summing over $p = 1, \dots, 9$ gives the multiplier $9 + 8 + 7 + \dots + 1 = 45$:

$$m = (9 + 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1) \times 8! = 45 \times 40\,320 = \boxed{1\,814\,400}.$$

Question 12

[Maximum mark: 5]

Calculator

Difficulty: Medium

A music company announces that they are going to release a list of their top 10 artists. If 5 of them are pop artists, 3 of them are rock artists and 2 of them are hip-hop artists, find the total number of possible ways in which

- (a) the rock artists are listed consecutively; [2]
- (b) the rock artists and the hip-hop artists are both listed consecutively, with the 3 rock artists listed immediately below the 2 hip-hop artists. [3]

Question 12

Solution

(a) Treat the 3 rock artists as a single block. This block plus the 5 pop artists and 2 hip-hop artists gives $5 + 1 + 2 = 8$ items to arrange, with the block internally orderable in $3!$ ways:

$$n = 8! \times 3! = 40\,320 \times 6 = \boxed{241\,920}.$$

(b) Now the 2 hip-hop artists (block H , orderable in $2!$ ways) sit immediately above the 3 rock artists (block R , orderable in $3!$ ways), so H and R form one combined block. This combined block plus the 5 pop artists gives 6 items to arrange:

$$m = 6! \times 2! \times 3! = 720 \times 2 \times 6 = \boxed{8\,640}.$$

Question 13

[Maximum mark: 8]

Calculator

Difficulty: Medium

A farmer has six sheep pens, arranged in a grid with three rows and two columns. Five sheep called Amber, Brownie, Curly, Daisy and Eden are to be placed in the pens. Each pen is large enough to hold all of the sheep. Amber and Brownie are known to fight. Find the number of ways of placing the sheep in the pens in each of the following cases:

- (a) Each pen is large enough to contain five sheep. Amber and Brownie must not be placed in the same pen. [4]
- (b) Each pen may only contain one sheep. Amber and Brownie must not be placed in pens which share a boundary. [4]

Question 13

Solution

(a) Each of the 5 sheep is independently placed into one of the 6 pens, so without restriction there are 6^5 ways. Subtract the cases where Amber and Brownie share a pen: Amber and Brownie together have 6 choices of shared pen, and the other 3 sheep are placed freely in 6^3 ways, giving 6×6^3 “same pen” outcomes.

$$6^5 - 6 \times 6^3 = 6^3 (6^2 - 6) = 216 \times 30 = \boxed{6480}.$$

(b) Now each pen holds exactly one sheep, so all $5!$ arrangements of the 5 sheep into 5 of the 6 pens are equally likely — but with only 5 sheep and 6 pens, one pen is always empty, so really we are placing 5 distinct sheep into 6 distinct pens with exactly one pen empty: there are $6 \times 5! = 720$ such arrangements in total (choose the empty pen in 6 ways, then arrange the 5 sheep in the remaining 5 pens in $5!$ ways).

In the 3×2 grid, a pen shares a boundary with 2 or 3 neighbouring pens (corner pens have 2 neighbours, the two pens in the middle row each have 3 neighbours). Counting ordered (Amber, Brownie) placements into boundary-sharing pens: each of the 4 corner pens contributes 2 neighbours, and each of the 2 middle pens contributes 3 neighbours, giving a total of $4 \times 2 + 2 \times 3 = 14$ ordered adjacent pen-pairs; with Amber and Brownie assigned to a specific ordered pair this is 14 ways, and the other 3 sheep then fill 3 of the remaining 4 pens (one pen stays empty) in $4 \times 3 \times 2 = 24$ ways.

$$14 \times 24 = 336$$

adjacent placements. Hence the number of valid arrangements is

$$720 - 336 = \boxed{384}.$$

Question 14

[Maximum mark: 6]

Calculator

Difficulty: Medium

Twelve students are to take an exam in advanced combinatorics. The exam room is set out in three rows of four desks, with the invigilator at the front of the room.

- (a) Find the number of ways the twelve students may be arranged in the exam hall. [1]

Two of the students, Helen and Nicky, are suspected of cheating in a previous exam.

- (b) Find the number of ways the students may be arranged if Helen and Nicky must sit so that one is directly behind the other (with no desk in between). [2]
- (c) Find the number of ways the students may be arranged if Helen and Nicky must not sit next to each other in the same row. [3]

Question 14

Solution

(a) The 12 students can be arranged in the 12 desks in

$$12! = \boxed{479\,001\,600}$$

ways.

(b) With three rows of four desks, each of the 8 desks in the front two rows has exactly one desk directly behind it, giving 8 front-back desk pairs, and Helen/Nicky can occupy either desk of a chosen pair in $2!$ ways: $8 \times 2 = 16$ placements. The remaining 10 students fill the remaining 10 desks in $10!$ ways:

$$16 \times 10! = \boxed{58\,060\,800}.$$

(c) There are 3 pairs of adjacent desks within each of the 3 rows of 4 desks, giving $3 \times 3 = 9$ adjacent desk-pairs in total, and Helen/Nicky occupy either desk of a chosen pair in $2!$ ways: $9 \times 2 = 18$ placements where they *are* next to each other in the same row. The remaining 10 students fill the rest in $10!$ ways, so

$$18 \times 10! = 65\,318\,400$$

arrangements have them adjacent in a row. Hence the number where they are **not** adjacent in the same row is

$$12! - 18 \times 10! = 479\,001\,600 - 65\,318\,400 = \boxed{413\,683\,200}.$$

Question 15

[Maximum mark: 8]

Calculator

Difficulty: Medium

Eight boys and two girls sit on a bench. Determine the number of possible arrangements, given that

- (a) the girls do not sit together; [3]
- (b) the girls do not sit on either end; [2]
- (c) the girls do not sit on either end and do not sit together. [3]

Question 15

Solution

(a) Let k be the number of arrangements where the two girls do **not** sit together. Subtract, from the total $10!$ arrangements, those where the girls *do* sit together (treat them as one block: $9!$ arrangements of 9 items, times $2!$ internal orderings):

$$k = 10! - 2 \times 9! = 3\,628\,800 - 725\,760 = \boxed{2\,903\,040}.$$

(b) Let m be the number of arrangements where the girls do not sit on either end. Choose which of the 8 boys sits on each end (8 then 7 choices), then arrange the remaining 8 people (2 girls + 6 boys) in the middle 8 places in $8!$ ways:

$$m = 8 \times 7 \times 8! = 56 \times 40\,320 = \boxed{2\,257\,920}.$$

(c) Let n be the number where the girls neither sit together nor on either end. Using the same end-selection (8×7 ways) but now subtracting, from the $8!$ middle arrangements, the $2 \times 7!$ arrangements where the two girls *are* adjacent within the middle eight places:

$$n = 8 \times 7 \times (8! - 2 \times 7!) = 56 \times (40\,320 - 10\,080) = 56 \times 30\,240 = \boxed{1\,693\,440}.$$

Question 16

[Maximum mark: 7]

Calculator

Difficulty: Medium

At a gathering of 12 teachers, seven are male and five are female. A group of five of these teachers go out for a meal together. Determine the possible number of groups in each of the following situations:

- (a) There are more males than females in the group. [4]
- (b) Two of the teachers, Gary and Gerwyn, refuse to go out for a meal together. [3]

Question 16

Solution

(a) “More males than females” in a group of 5 means 5, 4 or 3 males (with the rest female). Summing these three cases:

$$\binom{7}{5} + \binom{7}{4} \binom{5}{1} + \binom{7}{3} \binom{5}{2} = 21 + 35 \times 5 + 35 \times 10 = 21 + 175 + 350 = \boxed{546}.$$

(b) The complement of “Gary and Gerwyn not both included” is “both Gary and Gerwyn included”. If both are in the group, the other 3 members are chosen from the remaining 10 teachers in $\binom{10}{3}$ ways. Subtracting from the total number of groups of 5 from 12 teachers,

$$\binom{12}{5} - \binom{10}{3} = 792 - 120 = \boxed{672}.$$

Question 17

[Maximum mark: 4]

Calculator

Difficulty: Medium

Consider the set of six-digit positive integers that can be formed from the digits 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9. Find the total number of six-digit positive integers that can be formed such that

- (a) the digits are distinct; [2]
- (b) the digits are distinct and are in increasing order. [2]

Question 17

Solution

(a) The leading digit cannot be 0, so it can be chosen in 9 ways; the remaining 5 digits are chosen in order from the 9 digits left (which now includes 0):

$$9 \times {}^9P_5 = 9 \times 9 \times 8 \times 7 \times 6 \times 5 = \boxed{136\,080}.$$

(b) If the digits must be distinct *and* arranged in increasing order, then *every* choice of 6 distinct digits corresponds to exactly one valid six-digit number (the digits sorted into increasing order) — and since the digits increase, the number automatically cannot start with 0 unless 0 is the only digit chosen, which is impossible for 6 digits. So we simply choose 6 of the 9 non-zero digits $\{1, \dots, 9\}$ (0 can never be included, since it would have to be the smallest digit and thus the leading digit):

$$\binom{9}{6} = \boxed{84}.$$

Question 18

[Maximum mark: 6]

Calculator

Difficulty: Medium

A junior baseball team consists of six boys and three girls. The team members are to be placed in a line to have their photograph taken.

- (a) In how many ways can the team members be placed if
- (i) there are no restrictions;
 - (ii) the girls must be placed next to each other? [3]
- (b) Five members of the team are selected to attend a baseball summer camp. Find the number of possible selections that contain at least two girls. [3]

Question 18

Solution

(a)(i) With no restrictions, the 9 team members are placed in a line in

$$9! = \boxed{362\,880}$$

ways.

(a)(ii) Treat the 3 girls as one block. This block plus the 6 boys gives 7 items to arrange, with the block internally orderable in $3!$ ways:

$$3! \times 7! = 6 \times 5\,040 = \boxed{30\,240}.$$

(b) “At least two girls” among the 5 selected means exactly 2 girls (and 3 boys) or exactly 3 girls (and 2 boys):

$$\binom{3}{2} \binom{6}{3} + \binom{3}{3} \binom{6}{2} = 3 \times 20 + 1 \times 15 = 60 + 15 = \boxed{75}.$$

Question 19

[Maximum mark: 7]

Calculator

Difficulty: Medium

Malik owns 15 recipe books which he bought while travelling.

- (a) Find the number of different ways of arranging the books in a line on a shelf. [1]

Malik's books were bought in three continents. Six books are from Asia, five from Europe and four from Africa. Malik decides to arrange the books on the shelf so that the books from each of the continents are grouped together.

- (b) Determine the number of different ways that Malik can do this. [3]

Malik chooses four books, all from the same continent.

- (c) Determine the number of different choices Malik could make. [3]

Question 19

Solution

(a) All 15 distinct books can be arranged in a line in

$$15! = \boxed{1\,307\,674\,368\,000 \approx 1.31 \times 10^{12}}$$

ways.

(b) Treat each continent's books as a single block: this gives 3 blocks to arrange along the shelf ($3!$ ways), and within each block, the books can be ordered internally ($6!$, $5!$ and $4!$ ways respectively):

$$3! \times 6! \times 5! \times 4! = 6 \times 720 \times 120 \times 24 = \boxed{12\,441\,600}.$$

(c) "All from the same continent" means all 4 chosen books come from Asia (6 available), Europe (5 available) or Africa (exactly 4 available, so there is only 1 way to choose all of them):

$$\binom{6}{4} + \binom{5}{4} + \binom{4}{4} = 15 + 5 + 1 = \boxed{21}.$$

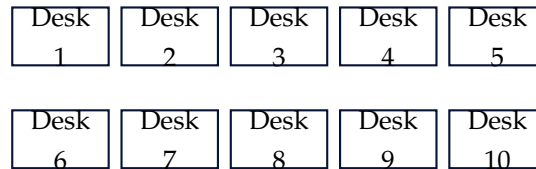
Question 20

[Maximum mark: 6]

Calculator

Difficulty: Hard

Ten students are to be arranged in a new chemistry lab. The chemistry lab is set out in two rows of five desks as shown below.



- (a) Find the number of ways the ten students may be arranged in the lab.
[1]

Two of the students, Hugo and Leo, were noticed to talk to each other during previous lab sessions.

- (b) Find the number of ways the students may be arranged if Hugo and Leo must sit so that one is directly behind the other (for example, Desk 1 and Desk 6). [2]
- (c) Find the number of ways the students may be arranged if Hugo and Leo must **not** sit next to each other in the same row. [3]

Question 20

Solution

(a) The 10 students can be arranged in the 10 desks in

$$10! = \boxed{3\,628\,800}$$

ways.

(b) There are 5 front-back desk pairs (Desk i , Desk $i + 5$), and Hugo/Leo can occupy either desk of a chosen pair in $2!$ ways, giving $5 \times 2 = 10$ placements. The remaining 8 students fill the remaining 8 desks in $8!$ ways. The number of arrangements is

$$10 \times 8! = 10 \times 40\,320 = \boxed{403\,200}.$$

(c) There are 8 pairs of adjacent desks within a row (4 in each row), and Hugo/Leo can occupy either desk of a chosen adjacent pair in $2!$ ways, giving $8 \times 2 = 16$ placements where they *do* sit next to each other in the same row. The remaining 8 students fill the rest in $8!$ ways, so

$$16 \times 8! = 16 \times 40\,320 = 645\,120$$

arrangements have Hugo and Leo adjacent in a row. Hence the number where they are **not** adjacent in the same row is

$$10! - 16 \times 8! = 3\,628\,800 - 645\,120 = \boxed{2\,983\,680}.$$

Question 21

[Maximum mark: 6]

Calculator

Difficulty: Hard

There are 11 players on a football team who are asked to line up in one straight line for a team photo. Three of the team members named Adam, Brad and Chris refuse to stand next to each other (no two of the three may be adjacent). There is no restriction on the order in which the other team members position themselves. Find the number of different orders in which the 11 team members can be positioned for the photo.

Question 21

Solution

Let A, B, C be the events “Adam & Brad adjacent”, “Adam & Chris adjacent”, “Brad & Chris adjacent” respectively. We want $11! - |A \cup B \cup C|$, found by inclusion–exclusion.

Single events: treating the named pair as one block gives 10 items to arrange, with $2!$ internal orderings:

$$|A| = |B| = |C| = 2 \times 10! = 7\,257\,600 \quad \Rightarrow \quad |A| + |B| + |C| = 21\,772\,800.$$

Pairwise intersections: e.g. $A \cap B$ means Adam is adjacent to *both* Brad and Chris, which forces the block *Brad–Adam–Chris* or *Chris–Adam–Brad* (2 orderings), arranged with the other 8 members as 9 items:

$$|A \cap B| = |A \cap C| = |B \cap C| = 2 \times 9! = 725\,760 \quad \Rightarrow \quad \text{sum} = 2\,177\,280.$$

Triple intersection: $|A \cap B \cap C| = 0$, since three people in a line who form a consecutive block of 3 only satisfy *two* of the three pairwise-adjacency conditions — the two people at the outer ends of the block are never adjacent to each other.

By inclusion–exclusion,

$$|A \cup B \cup C| = 21\,772\,800 - 2\,177\,280 + 0 = 19\,595\,520.$$

Hence the number of valid arrangements is

$$11! - 19\,595\,520 = 39\,916\,800 - 19\,595\,520 = \boxed{20\,321\,280}.$$

Question 21

Alternative Solution (Method of Gaps)

First arrange the other 8 team members in a line: this can be done in $8!$ ways. Doing so creates 9 gaps (before, between and after these 8 players) into which Adam, Brad and Chris can be placed so that no two of them are adjacent:

$$\square _ \square _ \square _ \cdots _ \square \quad (8 \text{ players, } 9 \text{ gaps}).$$

Choosing 3 of these 9 gaps, in order, for Adam, Brad and Chris (a different gap each, since two people in the same gap would be adjacent) can be done in $9 \times 8 \times 7$ ways. By the multiplication principle, the total number of arrangements is

$$8! \times (9 \times 8 \times 7) = 40\,320 \times 504.$$

$$= \boxed{20\,321\,280}.$$

Question 22

[Maximum mark: 6]

Calculator

Difficulty: Hard

The barcode strings of a new product are created from four letters A, B, C, D and ten digits $0, 1, 2, \dots, 9$. No three of the letters may be written consecutively in a barcode string. There is no restriction on the order in which the numbers can be written. Find the number of different barcode strings that can be created.

Question 22

Solution

First arrange the 10 digits: this can be done in $10!$ ways. This creates 11 gaps (before, between and after the digits) into which the 4 letters can be placed. Since no three letters may be consecutive, each gap may hold **at most 2** letters. There are three possible ways to distribute the 4 distinct letters among the gaps:

Case 1 — all 4 letters in different gaps: choose 4 of the 11 gaps and assign the 4 distinct letters to them:

$$\binom{11}{4} \times 4! = 330 \times 24 = 7\,920.$$

Case 2 — one gap holds 2 letters, the other 2 letters are alone in two further gaps: choose which 2 of the 4 letters are paired ($\binom{4}{2} = 6$ ways, ordered internally in $2!$ ways), then place this pair-block and the 2 remaining single letters into 3 of the 11 gaps, in order:

$$\binom{4}{2} \times 2! \times (11 \times 10 \times 9) = 6 \times 2 \times 990 = 11\,880.$$

Case 3 — two gaps each hold 2 letters: choose 2 of the 11 gaps for the pairs ($\binom{11}{2} = 55$ ways), split the 4 letters into an ordered pair-of-pairs for these two (distinguishable) gaps ($\binom{4}{2} = 6$ ways), and order each pair internally ($2! \times 2!$ ways):

$$\binom{11}{2} \times \binom{4}{2} \times 2! \times 2! = 55 \times 6 \times 4 = 1\,320.$$

Adding the three cases, the number of ways to place the letters is

$$7\,920 + 11\,880 + 1\,320 = 21\,120.$$

By the multiplication principle, the total number of barcode strings is

$$\begin{aligned} 10! \times 21\,120 &= 3\,628\,800 \times 21\,120 \\ &= \boxed{76\,640\,256\,000 \approx 7.66 \times 10^{10}}. \end{aligned}$$

Question 23

[Maximum mark: 6]

Calculator

Difficulty: Hard

In a trial examination session a candidate at a school has to take 18 examination papers including the physics paper, the chemistry paper and the biology paper. No two of these three papers may be taken consecutively. There is no restriction on the order in which the other examination papers may be taken. Find the number of different orders in which these 18 examination papers may be taken.

Question 23

Solution

Let X , Y , Z be the events “physics & chemistry adjacent”, “physics & biology adjacent” and “chemistry & biology adjacent” respectively. We want $18! - |X \cup Y \cup Z|$, found by inclusion–exclusion.

Single events: treating the named pair as one block gives 17 items to arrange, with $2!$ internal orderings:

$$|X| = |Y| = |Z| = 2 \times 17! \quad \Rightarrow \quad |X| + |Y| + |Z| = 3 \times 2 \times 17! = 6 \times 17!.$$

Pairwise intersections: e.g. $X \cap Y$ means physics is adjacent to *both* chemistry (from X) and biology (from Y), so physics is the *common* subject and must sit in the middle of the three, forcing the block *chemistry–physics–biology* or *biology–physics–chemistry* (2 orderings), together with the other 15 papers giving 16 items:

$$|X \cap Y| = 2 \times 16!.$$

Similarly, $X \cap Z$ forces chemistry into the middle (2 orderings) and $Y \cap Z$ forces biology into the middle (2 orderings), so

$$|X \cap Y| = |X \cap Z| = |Y \cap Z| = 2 \times 16! \quad \Rightarrow \quad \text{sum} = 3 \times 2 \times 16! = 6 \times 16!.$$

Triple intersection: $X \cap Y \cap Z$ would require *every* pair of the three subjects to be mutually adjacent at once. But once all three papers are placed consecutively (in any of the $3!$ orders), only the *two* pairs involving the middle paper are actually adjacent — the two “outer” papers of the block are never adjacent to each other. So it is impossible for all three pairwise-adjacency conditions to hold simultaneously (exactly the same subtlety as in Question 21’s football-team problem):

$$|X \cap Y \cap Z| = 0.$$

By inclusion–exclusion,

$$|X \cup Y \cup Z| = 6 \times 17! - 6 \times 16! + 0 = 6 \times 17! - 6 \times 16!.$$

Hence the number of valid orderings is

$$18! - (6 \times 17! - 6 \times 16!) = \boxed{4\,393\,785\,876\,480\,000 \approx 4.39 \times 10^{15}}.$$

Question 24

[Maximum mark: 7]

Calculator

Difficulty: Hard

Mary, three female friends, and her brother, Peter, attend the theatre. In the theatre there is a row of 10 empty seats. For the first half of the show, they decide to sit next to each other in this row.

- (a) Find the number of ways these five people can be seated in this row. [3]

For the second half of the show, they return to the same row of 10 empty seats. The four girls decide to sit at least one seat apart from Peter. The four girls do not have to sit next to each other.

- (b) Find the number of ways these five people can now be seated in this row. [4]

Question 24

Solution

(a) Treat the 5 people as one consecutive block occupying 5 of the 10 seats. The block can start at any of $10 - 5 + 1 = 6$ positions, and the 5 people are ordered within it in $5!$ ways:

$$6 \times 5! = \boxed{720}.$$

(b) Here only 5 of the 10 seats are occupied (the other 5 stay empty), and each of the 4 girls must *not* be adjacent to Peter's seat. Split into cases by Peter's seat:

Case 1 — Peter sits at either end (seats 1 or 10): only 1 neighbouring seat is forbidden, leaving $9 - 1 = 8$ allowed seats for the girls. Choosing and ordering 4 of these 8 seats: $8 \times 7 \times 6 \times 5 = {}^8P_4 = 1\,680$ ways, and Peter has 2 choices of end seat:

$$2 \times 1\,680 = 3\,360.$$

Case 2 — Peter sits in one of the 8 interior seats (seats 2–9): 2 neighbouring seats are forbidden, leaving $9 - 2 = 7$ allowed seats for the girls. Choosing and ordering 4 of these 7 seats: $7 \times 6 \times 5 \times 4 = {}^7P_4 = 840$ ways, and Peter has 8 choices of interior seat:

$$8 \times 840 = 6\,720.$$

Adding the two cases,

$$3\,360 + 6\,720 = \boxed{10\,080}.$$

Question 25

[Maximum mark: 5]

Calculator

Difficulty: Hard

Let S be the set of 30 positive integers $\{1, 2, 3, \dots, 28, 29, 30\}$. Raghu randomly selects three positive integers from S without replacement. He then adds them together and determines whether the sum is divisible by 3. Determine the total number of selections Raghu can make to obtain a sum that is divisible by 3. (Order is not important, so $\{1, 2, 3\}$ and $\{3, 2, 1\}$ are considered to be the same selection.) [5]

Question 25

Solution

Split S into three residue classes modulo 3: there are 10 numbers of the form $3n$ (i.e. $\equiv 0$), 10 of the form $3n - 1$ (i.e. $\equiv 2$) and 10 of the form $3n - 2$ (i.e. $\equiv 1$). A sum of three numbers is divisible by 3 exactly when the three residues sum to a multiple of 3, which happens in two distinct ways:

Case 1 — all three numbers from the *same* residue class: choose all 3 from one of the three classes of 10:

$$\binom{10}{3} + \binom{10}{3} + \binom{10}{3} = 3 \times 120 = 360.$$

Case 2 — one number from *each* of the three residue classes: choose 1 from each class of 10:

$$\binom{10}{1} \binom{10}{1} \binom{10}{1} = 10^3 = 1\,000.$$

Adding the two cases, the total number of valid selections is

$$360 + 1\,000 = \boxed{1\,360}.$$

Question 26

[Maximum mark: 7]

Calculator

Difficulty: Hard

A group of 10 children includes one pair of brothers, Alvin and Bobby, and one pair of sisters, Catalina and Daniela. The children are to be seated at 10 desks which are arranged in two rows of five. Alvin and Bobby must be seated next to each other in the same row.

- (a) Find the total number of ways the children can be seated. [3]

After an argument, Catalina and Daniela must not be seated next to each other. Alvin and Bobby must still be seated next to each other.

- (b) Find the total number of ways the children can be seated. [4]

Question 26

Solution

(a) In each row of 5 desks there are 4 pairs of adjacent desks, so across the two rows there are $2 \times 4 = 8$ adjacent desk-pairs. Alvin and Bobby occupy either desk of a chosen pair in $2!$ ways, giving $8 \times 2 = 16$ placements. The remaining 8 children fill the remaining 8 desks in $8!$ ways:

$$16 \times 8! = \boxed{645\,120}.$$

(b) From part (a), the number of seatings with Alvin & Bobby together (regardless of Catalina/Daniela) is 645 120. Subtract those where Catalina and Daniela are *also* adjacent to each other, splitting by *which* adjacent desk-pair Alvin and Bobby occupy.

In a row of 5 desks, the 4 adjacent desk-pairs are (1,2), (2,3), (3,4), (4,5): the two pairs (1,2) and (4,5) touch the **end** of the row, while (2,3) and (3,4) do **not**. Across both rows there are therefore 4 “end” pairs and 4 “non-end” pairs (8 in total, matching part (a)).

Case 1 — A&B occupy an end pair (e.g. desks 1–2): this leaves 3 desks in the same row (here, desks 3, 4, 5) and all 5 desks of the other row untouched. Among these 8 remaining desks, the adjacent pairs available for C&D are: 2 within the partial row (here, (3,4) and (4,5)) plus 4 within the full other row — $2 + 4 = 6$ pairs, each orderable in $2!$ ways, giving $6 \times 2 = 12$ ways to seat C&D together. With 4 choices of end-pair for A&B (each with $2!$ internal order, i.e. $4 \times 2 = 8$ placements) and the remaining 6 children filling the rest in $6!$ ways:

$$8 \times 12 \times 6! = 8 \times 12 \times 720 = 69\,120.$$

Case 2 — A&B occupy a non-end pair (e.g. desks 2–3): this leaves 3 desks in the same row (here, desks 1, 4, 5) and the other full row untouched. Among these 8 remaining desks, the adjacent pairs available for C&D

are: only 1 within the partial row (here, (4,5); desk 1 is now isolated from the rest) plus 4 within the full other row — $1 + 4 = 5$ pairs, giving $5 \times 2 = 10$ ways to seat C&D together. With 4 choices of non-end pair for A&B ($4 \times 2 = 8$ placements) and $6!$ ways for the rest:

$$8 \times 10 \times 6! = 8 \times 10 \times 720 = 57\,600.$$

So the number of seatings with *both* pairs together is $69\,120 + 57\,600 = 126\,720$. Hence the number with A&B together but C&D *not* together is

$$645\,120 - 126\,720 = \boxed{518\,400}.$$

Question 27

[Maximum mark: 6]

Calculator

Difficulty: Hard

A self-service sushi restaurant has a row of 12 available seats. Anvi, Vanya and Parita decide to go to the restaurant for lunch.

- (a) Find the number of possible ways that they can be seated in this row, if they decide **not** to sit together as a group of 3. [3]

The next day, Anvi, Vanya and Parita are joined by 3 additional people in the same restaurant, and they sit in the same row of 12 available seats. Anvi, Vanya and Parita now decide to sit next to each other as a group of 3.

- (b) Find the number of possible ways that these 6 people can be seated. [3]

Question 27

Solution

(a) With only 3 people seated among the 12 distinct seats (order matters, since the seats are distinguishable positions), the total number of ways to seat them is ${}^{12}P_3 = 12 \times 11 \times 10 = 1\,320$. Subtract the arrangements where all three sit together as a consecutive block of 3: the block can start at any of $12 - 3 + 1 = 10$ positions, ordered internally in $3!$ ways, giving $10 \times 3! = 60$ such arrangements. Hence

$$1\,320 - 60 = \boxed{1\,260}.$$

(b) Treat Anvi, Vanya and Parita as one block (orderable internally in $3!$ ways), which can start at any of the same 10 positions found above: $10 \times 3! = 60$ ways to place the block. The remaining 3 people are then seated, in order, in 3 of the remaining 9 seats: ${}^9P_3 = 9 \times 8 \times 7 = 504$ ways. By the multiplication principle,

$$60 \times 504 = \boxed{30\,240}.$$

Question 28

[Maximum mark: 5]

Calculator

Difficulty: Hard

A class of students plays a tic-tac-toe competition among themselves. Each individual game in the competition involves only two students. Every student in the class is to play every other student twice. However, Stephen left the class after he had played only seven games. All other games, not involving Stephen, were played. By the end of the competition a total of 513 games had been played. Determine the number of students that were originally in the class.

[5]

Question 28

Solution

Let n be the original number of students, so there are $n - 1$ students other than Stephen. If every pair of the $n - 1$ other students plays twice, that accounts for

$$2 \binom{n-1}{2} = (n-1)(n-2)$$

games not involving Stephen, and all of these were played. Adding Stephen's 7 games, the total number of games played is

$$(n-1)(n-2) + 7.$$

Setting this equal to 513,

$$(n-1)(n-2) + 7 = 513 \quad \Rightarrow \quad (n-1)(n-2) = 506.$$

Solving (e.g. by inspection or the quadratic formula), $n - 1 = 23$ and $n - 2 = 22$ satisfy $23 \times 22 = 506$, so $n - 1 = 23$, giving

$$n = \boxed{24}.$$